

Ninghan Zhong

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Google Scholar: <https://scholar.google.ca/citations?user=8zyHdjoAAAAJ&hl=en&oi=ao>

Email: ivan.nzhong@gmail.com

EDUCATION

B.S. in Computer Science | **University of Illinois at Urbana-Champaign**

Aug 2018 - May 2022

Graduated with the Highest Honors

MSc in Electrical and Computer Engineering | **University of Waterloo**

Aug 2023 - Aug 2025

Advised by Prof. Stephen Smith

Ph.D. in Robotics | **Georgia Institute of Technology**

Aug 2025 - Present

Advised by Prof. Sriram Vishwanath

Research Interests

Robot learning, generalizable manipulation, motion planning, robust physical interactions, representation learning

SELECTED PUBLICATIONS

- “Autonomous Navigation in Ice-Covered Waters with Learned Predictions on Ship-Ice Interactions”
 - **N. Zhong**, A. Potenza, and S. L. Smith
 - **ICRA 2025** [[paper](#)][[project page](#)]
- “Attentiveness Map Estimation for Haptic Teleoperation of Mobile Robot Obstacle Avoidance and Approach”
 - **N. Zhong** and K. Hauser
 - **IEEE RA-L 2024** [[paper](#)]
- “Bench-Push: Benchmarking Pushing-based Navigation and Manipulation Tasks for Mobile Robots”
 - **N. Zhong**, S. Caro, M. Ramesh, R. Bhatnagar, A. Iskandar, S. L. Smith
 - **CRV 2026** [[paper](#)]
- “AUTO-IceNAV: A Local Navigation Strategy for Autonomous Surface Ships in Broken Ice Fields”
 - R. D. Schaetzen, A. Botros, **N. Zhong**, K. Murrant, R. Gash, S. L. Smith
 - **IEEE T-RO 2025** [[paper](#)]
- “Hierarchical Intention Tracking for Robust Human-Robot Collaboration in Industrial Assembly Tasks”
 - Z. Huang*, Y. J. Mun*, X. Li†, Y. Xie†, **N. Zhong**†, W. Liang, J. Geng, T. Chen, and K. Driggs-Campbell
 - **ICRA 2023** [[paper](#)]

RESEARCH

Semantic Guided Vision-Language-Action Models for Generalizable Manipulation | **VLAs, steerability**

Advisor: Prof. Sriram Vishwanath

Department of Electrical and Computer Engineering, GaTech

- Conducting research on generalizable VLA architectural and training pipeline designs
- Developed steerable VLA framework that integrates conditional trace generation and semantic guidance
- Trained and evaluated the proposed framework both in simulation and on a physical YAM robotic arm

Benchmarking Pushing-based Interactive Navigation | **reinforcement learning, motion planning**

Advisor: Prof. Stephen L. Smith

Department of Electrical and Computer Engineering, UWaterloo

- Developed a suite of simulation benchmarks for pushing-based interactive navigation in cluttered settings
- Trained and evaluated multiple state-of-the-art learning-based policies in interactive navigation
- Work published in CRV 2026 [[paper](#)]

Predictive Path Planning for ASV Navigation | machine learning, planning, autonomous navigations

- Proposed a deep learning framework that predicts the coarse dynamics of ice obstacles for ASV navigations
- Presented a novel approach that integrates the learned model into a graph-search planner
- Evaluated both in simulation and a dry-land navigation testbed, outperforming existing methods
- Work published in ICRA 2025 [[paper](#)]

Attentiveness Estimation for Mobile Robot Teleoperation | human modeling, obstacle avoidance, haptics

Advisor: Prof. Kris Hauser

Department of Computer Science, UIUC

- Proposed a real-time human spatial attentiveness estimation model by combining visual saliency detection with computational working memory theories to optimize haptic feedback for mobile robot teleoperation
- Proposed an integrated robot perception pipeline that performs simultaneous environment mapping and human attentiveness estimation
- Experiments showed reduced task completion time, human control effort and obstacle collisions
- Work published in IEEE RA-L 2024 [[paper](#)]

Intention-Aware Human-Robot Collaboration | human-intention predictions, robot perception

Advisor: Prof. Katherine Driggs-Campbell

Electrical and Computer Engineering Department, UIUC

- Assisted with developing a robot control framework with hierarchical human-intention predictions for industrial assembly tasks
- Implemented an algorithm to localize target assembly parts in the robot workspace based on vision input from the robot wrist camera
- Work published in ICRA 2023 [[paper](#)]

Employment

Golden Ridge Robotics | Computer Vision & Algo. Intern - C++, ROS, OpenCV, PCL May 2021 - Jul 2021

- Developed an obstacle detection and localization model for low-speed sightseeing autonomous vehicles, achieving <30% relative error in obstacle dimension and <2% relative error in distance estimations
- Developed a 3D autonomous navigation environment marking program that allows marking static obstacles and driving areas with arbitrary shapes in point clouds for up to 25 km² with a decimeter-level accuracy

HONORS

- International Master's Award of Excellence \$12,500 CAD University of Waterloo, Nov 2023
- The Highest Honors at graduation UIUC, Spring 2022
- UIUC Dean's List (7 Semesters) UIUC, Fall 2018 - Spring 2020, Spring 2021 - Spring 2022
- James Scholars (2 Semesters) UIUC, Fall 2021, Spring 2022

Journal and Conference Reviewers

- IEEE Transactions on Robotics (T-RO)
- Reliability Engineering & System Safety (RESS)
- 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- 2025 IEEE International Symposium on Multi-Robot & Multi-Agent Systems (MRS)